YAZAN ELTOPGEY

Mechatronics Engineer

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Mechatronics Engineer with expertise in autonomous robotics, AI, and industrial automation. Proficient in ROS2, reinforcement learning, and computer vision. Published researcher with proven success in award-winning autonomous systems.

Experience

Autonomous Service Robot Research | Graduation Project *2024-June 2025*

- **DEVELOPED** hybrid navigation system combining reinforcement learning with traditional algorithms
- **IMPLEMENTED** full sensor fusion (camera, LiDAR, encoders) with SLAM mapping on Raspberry Pi
- **AWARDED** National Winner for Best Graduation Project in Robotics and AI (2025)

Robotics Research Intern | India *Jul 2024 - Oct 2024*

- **PUBLISHED** research paper: "Autonomous Mobile Robot Navigation using RL with Nav2
- **OPTIMIZED** ROS2 navigation stack reducing path planning computation time by 25%
- **DEVELOPED** reinforcement learning algorithms for dynamic environment navigation

Mental Health Prediction Research | Independent Project *2023-2024*

- **DEVELOPED** machine learning models for depression and anxiety prediction achieving 85% accuracy
- **IMPLEMENTED** KNN and SVM algorithms for early mental health disorder detection
- **CONDUCTED** comparative analysis of supervised learning techniques for healthcare applications

Industrial IoT & Automation Intern | INJO4.0, Jordan *Aug 2023 – Jan 2024*

- **IMPLEMENTED** predictive maintenance systems reducing equipment downtime by 40%
- **AUTOMATED** data collection workflows processing 5000+ daily sensor readings
- **DEVELOPED** real-time monitoring dashboards using PLCs and Norvi IIoT devices

Education

Bachelor of Science in Mechatronics Engineering *September 2025*
The Hashemite University, Zarqa, Jordan

Skills

Programming: C++, Python, CUDA, MATLAB
Robotics: ROS, ROS2, Gazebo, Isaac Sim, SLAM
AI/ML: Machine Learning, Deep Learning, Reinforcement Learning, OpenCV, TensorFlow, PyTorch, YOLO
Hardware: Arduino, Raspberry Pi, Jetson Nano,esp32,Stm32
Industrial: PLC Programming, IoT, IIoT, SCADA
Design: SolidWorks, AutoCAD, Fusion 360, Blender
Tools: GitHub, Linux, Docker, n8n

PROJECT

- Autonomous Service Robot
- Autonomous Vehicle in CARLA Simulation
- Mobile Robot Navigation with RL
- Cancer Detection with Deep Learning
- Mental Health Prediction
- Industrial IoT & Automation
- IOT/IIOT System
- Sumo Robot Competition